ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q2, Q4

Question 1

(a) (i) $f_x = f/p = 16/5 =$ **3.2 px**.

(ii) Width of sensor = 1280 × 5 = 6400. FOV = $N \cdot Z/f_x$ = 1280 × 400 / 3.2 = **160000 mm**.

One pixel = FOV/N = 160000/1280 = **125 mm per pixel**. (looks large but the formula is right)

(b) Resolution of the ADC = $V_{range}/2^n$ = 5 V / 4096 = 1.22×10^{-3} V = **1.22 mV**.

So the smallest change the system can resolve is 1.22 mV.

(c) ZOH discretisation: $F = e^{AT}$, $G = \int_0^T e^{A\tau} B \, d\tau = A^{-1}(F - I)B$ ($A \neq 0$).

Here $A = -2$, $B = 1$, $T = 0.1$: $F = e^{-0.2} =$ **0.8187**. $G = (F - 1)/A = (0.8187 - 1)/(-2) =$ **0.0906**.

$x_{k+1} = 0.8187\, x_k + 0.0906\, u_k$.

Question 2

(a) $x_1 = \theta$, $x_2 = \theta^{\cdot}$. $x^{\cdot}_1 = x_2$, $x^{\cdot}_2 = (1/J)(\tau - b\, x_2 - mg\ell \sin x_1)$.

Equilibrum means $x^{\cdot} = 0$, so $x_2 = 0$ and $mg\ell \sin \theta_e = \tau_0$. Sence sine is periodic we can always find a θ_e that satisfies this, so an equilibrium exists for any τ_0.

(b) Jacobian at $\theta_e = 60°$: $A = [0,\ 1;\ -(mg\ell/J)\cos\theta_e,\ -b/J]$.

$mg\ell/J = 2.4525/0.02 = 122.625$; $\cos(60) = -0.9524$, so $A_{21} = -122.625 \times (-0.9524) = +116.8$. $A_{22} = -b/J = -2.5$.

A = [0, 1; 116.8, −2.5], **B = [0; 50]**.

(c) At rest $\theta^{\cdot} = \theta^{\cdot\cdot} = 0$, so $K_p(\theta_d - \theta_{ss}) = mg\ell \sin \theta_{ss}$, i.e. $50(90 - \theta_{ss}) = 2.4525 \sin \theta_{ss}$.

Gravity torque is non-zero at 90°, so the left side can't be zero, hence $\theta_{ss} \neq \theta_d$.

Taking $\sin \theta_{ss} \approx 1$: $90 - \theta_{ss} = 2.4525/50 = 0.049$, so the stedy-state error ≈ **0.049°**.

(d) Error: $K_p(\theta_d - \theta_{ss}) = 0$ at rest, so $\theta_{ss} = \theta_d$; **error = 0**.

Poles: yes they change. The gravity term is removed from the dynamics so the effective stiffness changes from $K_p + mg\ell \cos\theta$ to K_p, which moves the poles (the system becomes faster).

Question 4

(a) Substituting u: $\theta^{\cdot\cdot} + (a + bk_2)\theta^{\cdot} + bk_1\theta = bN_{ff}\, r$ (the c term is the graviy load, handled separately).

So $s^2 + 4s + 2 = 0$, giving $s = -0.586$ and $s = -3.414$. DC gain = $bN_{ff}/(bk_1) = N_{ff}/2 = 1 \Rightarrow$ **$N_{ff} = 2$**.

(b) The actuator is 25% weaker, so the output is also 25% lower: $\theta_{ss} = 0.75$, and the tracking error = **25%**.

(c) Substitutting the control law and differentiating once: $s^3 + (c + bk_2)s^2 + (a + bk_1)s + bK_I$.

Desired: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

$6 + k_2 = 8 \Rightarrow$ **$k_2 = 2$**; $2 + k_1 = 24 \Rightarrow$ **$k_1 = 22$**; $K_I =$ **32**.

(d) For a constant r, in steady state all derivatives vanish: $\xi^{\cdot} = r - \theta_{ss} = 0 \Rightarrow \theta_{ss} = r$, independent of b and c; the closed loop only needs to be stable so that a stedy state exists.

Failure mode: integrater windup. When the acuator saturates the error persists, ξ keeps growing and causes a large overshoot afterwards. Remidy: anti-windup (clamp the integrator, conditional

integration or back-calculation).